P4.26 $\vec{S} = \frac{\hbar}{2}(\sigma_x, \sigma_y, \sigma_z)$

(a) $[S_x, S_y] = \frac{\hbar^2}{4}(\sigma_x\sigma_y - \sigma_y\sigma_x) = \frac{\hbar^2}{4}\left(\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}0&-i\\i&0\end{pmatrix} - \begin{pmatrix}0&-i\\i&0\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix}\right) =$

$= \frac{\hbar^2}{4}\left(\begin{pmatrix}i&0\\0&-i\end{pmatrix} - \begin{pmatrix}-i&0\\0&i\end{pmatrix}\right) = \frac{\hbar^2}{2} i \begin{pmatrix}1&0\\0&-1\end{pmatrix} = i\hbar\frac{\hbar}{2}\sigma_z = i\hbar S_z.$

the same for other possibilities

(b) $\sigma_j\sigma_k = \delta_{jk} + i\sum_\ell \varepsilon_{jk\ell}\sigma_\ell$ (4.153)

$\sigma_x\sigma_x = \begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix} = \begin{pmatrix}1&0\\0&1\end{pmatrix} = \begin{pmatrix}0&-i\\i&0\end{pmatrix}\begin{pmatrix}0&-i\\i&0\end{pmatrix} = \begin{pmatrix}1&0\\0&1\end{pmatrix}$

clearly $\sigma_z\sigma_z = \begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix} = \begin{pmatrix}1&0\\0&1\end{pmatrix}$ as well.

Thus for $j=k$: $\sigma_j\sigma_j = \delta_{jj} + i\sum_\ell \underbrace{\varepsilon_{jj\ell}}_{0 \text{ when any } \underline{\text{two}} \text{ indices coincide.}}\sigma_\ell = \delta_{jj} = 1.$

$\sigma_x\sigma_y = \begin{pmatrix}i&0\\0&-i\end{pmatrix} = i\sigma_z$

$\sigma_y\sigma_x = \begin{pmatrix}-i&0\\0&i\end{pmatrix} = -i\sigma_z$

$\sigma_y\sigma_z = \begin{pmatrix}0&-i\\i&0\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix} = \begin{pmatrix}0&i\\i&0\end{pmatrix} = i\sigma_x$

$\sigma_z\sigma_y = -i\sigma_x$

$\sigma_z\sigma_x = \begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix} = \begin{pmatrix}0&1\\-1&0\end{pmatrix} = i\begin{pmatrix}0&-i\\i&0\end{pmatrix} = i\sigma_y$

$\sigma_x\sigma_z = -i\sigma_y$

(Eq. 4.153) just collects all these identities in a single formula.

P4.27 $\chi = A \begin{pmatrix} 3i \\ 4 \end{pmatrix}$

(a) $\langle \chi | \chi \rangle = |A|^2 \, (-3i, 4) \begin{pmatrix} 3i \\ 4 \end{pmatrix} = |A|^2 (9+16) = 1$

$\Rightarrow A = \frac{1}{5}$

$\chi = \frac{1}{5} \begin{pmatrix} 3i \\ 4 \end{pmatrix}$

(b) $\langle S_x \rangle = \frac{\hbar}{2} \left( \frac{-3i}{5}, \frac{4}{5} \right) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{3i}{5} \\ \frac{4}{5} \end{pmatrix} = \frac{\hbar}{2} \left( \frac{-3i}{5}, \frac{4}{5} \right) \begin{pmatrix} \frac{4}{5} \\ \frac{3i}{5} \end{pmatrix} =$

$= \frac{\hbar}{2} \left( \frac{-12i}{25} + \frac{12i}{5} \right) = 0$

$\langle S_y \rangle = \frac{\hbar}{2} \left( -\frac{3i}{5}, \frac{4}{5} \right) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} \frac{3i}{5} \\ \frac{4}{5} \end{pmatrix} =$

$= \frac{\hbar}{2} \frac{1}{25} (-3i, 4) \begin{pmatrix} -4i \\ -3 \end{pmatrix} = \frac{\hbar}{2} \frac{1}{25} (-12 - 12) = -\frac{12}{25} \hbar$

$\langle S_z \rangle = \frac{\hbar}{2} \frac{1}{25} (-3i, 4) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 3i \\ 4 \end{pmatrix} = \frac{\hbar}{50} (-3i, 4) \begin{pmatrix} 3i \\ -4 \end{pmatrix} =$

$= \frac{\hbar}{50} (9 - 16) = -\frac{7}{50} \hbar .$

(c) Find standard deviations $\sigma_{S_x} = \langle S_x^2 \rangle - \langle S_x \rangle^2$ etc.

But $S_x^2 = S_y^2 = S_z^2 = \frac{\hbar^2}{4}$

$\Rightarrow \sigma_{S_x} = \sqrt{\frac{\hbar^2}{4} - 0} = \frac{\hbar}{2}$

$\sigma_{S_y}^2 = \frac{\hbar^2}{4} - \left( -\frac{12}{25} \hbar \right)^2 = \frac{\hbar^2}{2500} (625 - 576) = \frac{49}{2500} \hbar^2$

$\Rightarrow \sigma_{S_y} = \frac{7}{50} \hbar$

$\sigma_{S_z}^2 = \frac{\hbar^2}{4} - \left( \frac{7}{50} \right)^2 \hbar^2 = \frac{576}{2500} \hbar^2 \Rightarrow \sigma_{S_z} = \frac{12}{25} \hbar .$

(d) $\sigma_{S_x}\sigma_{S_y} = \frac{\hbar}{2}\cdot\frac{7}{50}\hbar$

while $\langle S_z\rangle = \frac{7}{50}\hbar$

From (4.100) $\sigma_{S_x}\sigma_{S_y} \geq \frac{\hbar}{2}\left|\langle S_z\rangle\right|$

$$\frac{\hbar}{2}\cdot\frac{7\hbar}{50} \geq \frac{\hbar}{2}\,\frac{7\hbar}{50} \qquad \underline{\text{ok.}}$$

$$\sigma_{S_y}\sigma_{S_z} = \frac{7\hbar}{50}\,\frac{12\hbar}{25} \geq \frac{\hbar}{2}\left|\langle S_x\rangle\right| = 0 \qquad \underline{\text{ok}}$$

$$\sigma_{S_z}\sigma_{S_x} = \frac{12\hbar}{25}\cdot\frac{\hbar}{2} \geq \frac{\hbar}{2}\left|\langle S_y\rangle\right| = \frac{\hbar}{2}\,\frac{12}{25}\hbar \qquad \underline{\text{ok}}$$

<u>P4.29</u> $\hat{S}_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

(a) $\frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \lambda\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$$\Rightarrow \det\begin{pmatrix} -\lambda & -i\frac{\hbar}{2} \\ i\frac{\hbar}{2} & -\lambda \end{pmatrix} = \lambda^2 - \left(i\frac{\hbar}{2}\right)\left(-i\frac{\hbar}{2}\right) = 0$$

$$\Rightarrow \lambda = \pm\frac{\hbar}{2}$$

Eigenspinors: $\frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \pm\frac{\hbar}{2}\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$$\begin{cases} -i\beta = \pm\alpha \Rightarrow \beta = \pm i\alpha \\ i\alpha = \pm\beta \end{cases}$$

Thus $\chi_+^{(y)} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}$, $\chi_-^{(y)} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$

$\left(\lambda_+ = +\frac{\hbar}{2}\right)$ $\left(\lambda_- = -\frac{\hbar}{2}\right)$

(b) $\chi = \begin{pmatrix} a \\ b \end{pmatrix} = c_1 \chi_+^{(y)} + c_2 \chi_-^{(y)}$

Find $c_{1,2}$ via: $c_1 = \langle \chi_+^{(y)} | \chi \rangle = \frac{1}{\sqrt{2}} (1, -i) \begin{pmatrix} a \\ b \end{pmatrix} =$

$= \frac{1}{\sqrt{2}} (a - ib)$ ;

$c_2 = \langle \chi_-^{(y)} | \chi \rangle = \frac{1}{\sqrt{2}} (1, i) \begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{\sqrt{2}} (a + ib)$

$\Rightarrow \chi = \begin{pmatrix} a \\ b \end{pmatrix} = \frac{a - ib}{\sqrt{2}} \chi_+^{(y)} + \frac{a + ib}{\sqrt{2}} \chi_-^{(y)}$

$\Rightarrow$ measurement of $S_y$ will give: $\begin{cases} +\frac{\hbar}{2} & \text{with prob. } p_+ = \frac{|a-ib|^2}{2} \\ -\frac{\hbar}{2} & p_- = \frac{|a+ib|^2}{2} \end{cases}$

$p_+ + p_- = \frac{|a-ib|^2}{2} + \frac{|a+ib|^2}{2} =$

$= \frac{1}{2} \left( (a - ib)(a^* + ib^*) + (a + ib)(a^* - ib^*) \right) =$

$= \frac{1}{2} \left( |a|^2 + |b|^2 \underline{- iba^*} \underline{+ iab^*} + |a|^2 + |b|^2 \underline{+ iba^*} \underline{- iab^*} \right)$

$= |a|^2 + |b|^2 = 1$ because $\langle \chi | \chi \rangle = 1$.

(c) $S_y^2 = \frac{\hbar^2}{4} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$\Rightarrow$ you get $S_y^2 = \frac{\hbar^2}{4}$ with probability 1.

P4.49 $\chi = A\begin{pmatrix}1-2i\\2\end{pmatrix}$

(a) $\langle\chi|\chi\rangle = |A|^2\,(1+2i,\,2)\begin{pmatrix}1-2i\\2\end{pmatrix} = |A|^2(1+4+4) = 9|A|^2 = 1$

$\Rightarrow A = \frac{1}{3}$

(b) Expand $\chi$ in the basis of $\chi_\pm^{(z)}$ eigenspinors:

$\chi = a\,\chi_+^{(z)} + b\,\chi_-^{(z)} \qquad \chi_+ = \begin{pmatrix}1\\0\end{pmatrix} \quad \chi_- = \begin{pmatrix}0\\1\end{pmatrix}$

$a = \langle\chi_+^{(z)}|\chi\rangle = (1,0)\,\frac{1}{3}\begin{pmatrix}1-2i\\2\end{pmatrix} = \frac{1-2i}{3}$

$b = \langle\chi_-^{(z)}|\chi\rangle = (0,1)\,\frac{1}{3}\begin{pmatrix}1-2i\\2\end{pmatrix} = \frac{2}{3}$

$\Rightarrow$ measurement of $S_z$ gives $\begin{cases} +\frac{\hbar}{2} \text{ with prob. } \left|\frac{1-2i}{3}\right|^2 = \frac{5}{9} \\ -\frac{\hbar}{2} \text{ with } p_- = \left|\frac{2}{3}\right|^2 = \frac{4}{9}\end{cases}$

(c) Expand $\chi$ in $\chi_\pm^{(x)}$ basis:

$\chi = c_1\,\chi_+^{(x)} + c_2\,\chi_-^{(x)} \quad ; \quad \chi_+^{(x)} = \frac{1}{\sqrt2}\begin{pmatrix}1\\1\end{pmatrix},\ \chi_-^{(x)} = \frac{1}{\sqrt2}\begin{pmatrix}1\\-1\end{pmatrix}$

$c_1 = \frac{1}{\sqrt2}(1,1)\,\frac{1}{3}\begin{pmatrix}1-2i\\2\end{pmatrix} = \frac{1}{3\sqrt2}(1-2i+2) = \frac{3-2i}{3\sqrt2}$

$c_2 = \frac{1}{\sqrt2}(1,-1)\,\frac{1}{3}\begin{pmatrix}1-2i\\2\end{pmatrix} = \frac{1}{3\sqrt2}(1-2i-2) = \frac{-1-2i}{3\sqrt2}$

$\Rightarrow S_x = \begin{cases} +\frac{\hbar}{2} \text{ with } |c_1|^2 = \frac{9+4}{9\cdot2} = \frac{13}{18} \\ -\frac{\hbar}{2} \text{ with } |c_2|^2 = \frac{1+4}{9\cdot2} = \frac{5}{18}\end{cases}$

(d) Expand $\chi$ in $\chi_\pm^{(y)}$ basis now: $\chi_\pm^{(y)} = \frac{1}{\sqrt2}\begin{pmatrix}1\\\pm i\end{pmatrix}$ (P4.29)

$\chi = r_1\,\chi_+^{(y)} + r_2\,\chi_-^{(y)}$

$r_1 = \langle\chi_+^{(y)}|\chi\rangle = \frac{1}{\sqrt2}(1\ \ i)\,\frac{1}{3}\begin{pmatrix}1-2i\\2\end{pmatrix} = \frac{1-2i-2i}{3\sqrt2} = \frac{1-4i}{3\sqrt2}$

$$r_2 = \langle \chi_-^{(y)} | \chi \rangle = \frac{1}{\sqrt{2}} (1, i) \frac{1}{3} \begin{pmatrix} 1-2i \\ 2 \end{pmatrix} = \frac{1-2i+2i}{3\sqrt{2}} = \frac{1}{3\sqrt{2}}$$

$$\Rightarrow S_y = \begin{cases} +\frac{\hbar}{2} & \text{with } p_+ = |r_1|^2 = \left(\frac{1-4i}{3\sqrt{2}}\right)^2 = \frac{1+16}{9 \cdot 2} = \frac{17}{18} \\ -\frac{\hbar}{2} & \text{with } p_- = |r_2|^2 = \left(\frac{1}{3\sqrt{2}}\right)^2 = \frac{1}{18} \end{cases}$$

Expectation values:

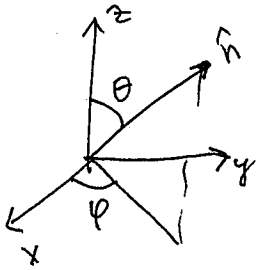
$$\langle S_y \rangle = +\frac{\hbar}{2} \cdot \frac{17}{18} + \left(-\frac{\hbar}{2}\right) \cdot \frac{1}{18} = \frac{\hbar}{2} \cdot \frac{16}{18} = \frac{4\hbar}{9}$$

$$\langle S_x \rangle = +\frac{\hbar}{2} \cdot \frac{13}{18} + \left(-\frac{\hbar}{2}\right) \cdot \frac{5}{18} = \frac{\hbar}{2} \cdot \frac{8}{18} = \frac{2\hbar}{9}$$

$$\langle S_z \rangle = +\frac{\hbar}{2} \cdot \frac{5}{9} + \left(-\frac{\hbar}{2}\right) \cdot \frac{4}{9} = \frac{\hbar}{2} \frac{1}{9} = \frac{\hbar}{18}$$

P4.30 $\hat{r} = \sin\theta\cos\varphi\,\hat{i} + \sin\theta\sin\varphi\,\hat{j} + \cos\theta\,\hat{k}$

$\hat{S}_r = \vec{S}\cdot\hat{r}$




$$S_r = \frac{\hbar}{2}\left(\sigma_x \sin\theta\cos\varphi + \sigma_y \sin\theta\sin\varphi + \sigma_z\cos\theta\right)$$

$$= \frac{\hbar}{2}\begin{pmatrix} \cos\theta & \sin\theta\, e^{-i\varphi} \\ \sin\theta\, e^{i\varphi} & -\cos\theta \end{pmatrix}$$

1) Eigenvalues: $\det\left(S_r - \frac{\hbar}{2}\lambda\right) = 0$

$$\Rightarrow \begin{pmatrix} \cos\theta - \lambda & \sin e^{-i\varphi} \\ \sin\theta\, e^{i\varphi} & -\cos\theta - \lambda \end{pmatrix} = \lambda^2 - \cos^2\theta - \sin^2\theta = 0$$

$$\Rightarrow \lambda = \pm 1$$

$\Rightarrow$ eigenvalues are $\pm\frac{\hbar}{2}$.

2) $$\frac{\hbar}{2}\begin{pmatrix} \cos\theta & \sin\theta\, e^{-i\varphi} \\ \sin\theta\, e^{i\varphi} & -\cos\theta \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \pm\frac{\hbar}{2}\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\begin{cases} \cos\theta\,\alpha + \sin\theta\, e^{-i\varphi}\beta = \pm\alpha \\ \sin\theta\, e^{i\varphi}\alpha - \cos\theta\,\beta = \pm\beta \end{cases}$$

$\oplus$ $$\underbrace{(1-\cos\theta)}_{2\sin^2\frac{\theta}{2}}\alpha = \underbrace{\sin\theta\, e^{-i\varphi}}_{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\beta \Rightarrow \sin\frac{\theta}{2}\,\alpha = \cos\frac{\theta}{2}\, e^{-i\varphi}\beta$$

$\ominus$ $$\underbrace{(-1-\cos\theta)}_{-2\cos^2\frac{\theta}{2}}\alpha = \underbrace{\sin\theta\, e^{-i\varphi}}_{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\beta \Rightarrow -\cos\frac{\theta}{2}\,\alpha = \sin\frac{\theta}{2}\, e^{-i\varphi}\beta$$

The spinor must be normalized $|\alpha|^2 + |\beta|^2 = 1$

$\oplus$ $$|\alpha|^2 + \left|\tan\frac{\theta}{2}\, e^{i\varphi}\alpha\right|^2 = 1 \Rightarrow |\alpha|^2\left(1 + \tan^2\frac{\theta}{2}\right) = \frac{|\alpha|^2}{\cos^2\frac{\theta}{2}} = 1$$

$$\Rightarrow \alpha = \cos\frac{\theta}{2} \Rightarrow \beta = \sin\frac{\theta}{2}\, e^{i\varphi} : \chi_+^{(+)} = \begin{pmatrix} \cos\theta/2 \\ \sin\theta/2\, e^{i\varphi} \end{pmatrix}$$

$$\ominus \quad |\alpha|^2 + \left| -\frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}} e^{i\varphi} \right|^2 |\alpha|^2 = 1$$

$$\Rightarrow \frac{|\alpha|^2}{\sin^2\frac{\theta}{2}} = 1 \Rightarrow \quad \alpha = \sin\theta/2 \qquad \beta = -\cos\frac{\theta}{2} e^{i\varphi}$$

$$\chi_-^{(r)} = \begin{pmatrix} \sin\theta/2 \\ -\cos\frac{\theta}{2} e^{i\varphi} \end{pmatrix}$$

Check: $\langle \chi_- | \chi_+ \rangle = \left(\sin\frac{\theta}{2}, -\cos\frac{\theta}{2} e^{-i\varphi}\right) \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2} e^{i\varphi} \end{pmatrix} =$

$$= \sin\frac{\theta}{2}\cos\frac{\theta}{2} - \cos\frac{\theta}{2}\sin\frac{\theta}{2} = 0.$$

<u>P4.56</u> (a) $f(\phi + \varphi_0) = e^{iL_z\varphi_0/\hbar} f(\phi)$

$$L_z = \frac{\hbar}{i}\frac{\partial}{\partial\phi} \Rightarrow e^{iL_z\varphi_0/\hbar} = e^{\varphi_0 \frac{\partial}{\partial\phi}} f(\phi)$$

For direction $\hat{n}$ : $\vec{L}\cdot\hat{n}/\hbar$ = generator of rotations about $\hat{n}$.

For $S = \frac{1}{2}$ $\quad \vec{L} = \frac{\hbar}{2}\vec{\sigma} \Rightarrow e^{i\vec{L}\cdot\hat{n}/\hbar\,\varphi_0} = e^{i\frac{\varphi_0}{2}\vec{\sigma}\cdot\hat{n}}$

$\Rightarrow$ <u>spinors</u> rotate as $\chi' = e^{i(\vec{\sigma}\cdot\hat{n})\varphi_0/2}\chi$

(b) $R_x(\pi) = e^{i\frac{\pi}{2}\hat{\sigma}_x} = \cos\frac{\pi}{2} + i\sigma_x \sin\frac{\pi}{2} = i\sigma_x = i\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$R_x(\pi)\chi_+ = i\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = i\begin{pmatrix} 0 \\ 1 \end{pmatrix} = i\chi_-$$

(c) $R_y(\pi/2) = e^{i\frac{\pi}{4}\sigma_y} = \cos\frac{\pi}{4} + i\sigma_y \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}(1 + i\sigma_y) =$

$$= \frac{1}{\sqrt{2}}\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

$$R_y(\pi/2)\chi_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}}(\chi_+ - \chi_-) = \chi_-^{(x)}$$